

Indoor Positioning System: Improved deep learning approach based on LSTM and multi-stage activity classification

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Abstract

Pedestrian dead reckoning (PDR) for indoor localization has the privilege of utility without prior environment knowledge and adjunct infrastructure. It can be a potentially scalable solution for various indoor positioning, visually impaired people navigation aide, and commercial applications. In this study, we develop a multi-stage deep learning-based approach to detect and estimate the stride and heading of a user. This approach takes advantage of classifying user action units from inertial sensors of smartphone and relevant action units and is then separately processed to estimate user displacement in terms of displacement distance and direction, respectively, with automatic feature extraction. The proposed system provides improved performance over the preceding deep learning model. It also exhibits two-dimensional finer resolution maneuvering of the user in contrast to only the left and right turn. Experiments were conducted to train and evaluate the proposed system's performance, the results of which validate the improved utility of our deep learning-based system.

Keywords: Indoor localization, smartphone IMU (inertial measurement unit) sensors, multi-stage, long short-term memory (LSTM).

1. Introduction

Indoor localization services hold commercial value estimation of 10 billion USD by 2020 [1]. The service application areas include healthcare monitoring, localized information delivery, context awareness, network security management, augmented reality, navigation for visually impaired persons, safety, and emergencies particularly [2, 3]. GPS navigation requires direct line-of-sight between users and satellites; also, GPS signals do not penetrate well through buildings. Hence the development of indoor localization techniques is an active research area. Various radio and non-radio based solutions have been proposed, but their applications are limited by

the deployment of external equipment and calibration of the indoor environment [2, 4]. On the other hand, pedestrian dead reckoning (PDR) is becoming increasingly prominent, taking advantage of inertial measurement unit (IMU) sensors of smartphones and other wearable devices; hence extra infrastructure is not required. The accelerometer, gyroscope, and magnetometer from inertial measurement unit (IMU) sensors record the motion information of the user wearing them to estimate the current position of the user with reference to starting position. This approach is constraint-free of the external environment, layout, and offline calibration [5]. Conventional methods including Kalman, Particle, and Complementary filters have been applied for feature extraction for inertial sensor-based PDR (IPDR) in recent studies. Gait motion and phase classification has been performed on extracted features using fixed or adaptive thresholding and machine learning methods. Whereas step and heading systems further estimate user trajectory using pre-set mathematical models, studies [5-7] provide detailed reviews of these IPDR methods. Deep learning-based step estimation methods using automatic feature extraction and selection have also been developed to overcome challenges of intra-user gait pattern variation and constrained way of holding smartphone devices [8-11]. However, they do not include the heading direction estimation of the user. Hussain et al. [12] developed deep learning-based IPDR indoor positioning system using long short-term memory (LSTM) units at two stages to classify user activity and action units from raw segmented signals. The said approach has limitations of heading classification in the left and right direction turns of the user only. Furthermore, it provides averaged step lengths of fixed small, medium, and large step sizes. This work presents multistage deep learning architecture where step size and heading of users are estimated separately. The proposed end-to-end trained network on user trails data using automatic feature extraction provides an estimated step length of each individual user stride in addition to the finer resolution of user heading direction.

2. Materials and methods

2.1. Preliminaries

Pedestrian dead reckoning (PDR) determines displacement and heading direction from the initial position to the user's final or current position. Using inertial measurement unit (IMU) sensors of a smartphone, displacement estimation can be attained from the accelerometer and gyroscope, while heading estimation can be obtained from the gyroscope, magnetometer, and accelerometer measurements during consecutive gait steps. Magnetometer measurements are susceptible to earth's magnetic field and external interference; hence gyroscope is typically used with compensation to its drift problem only. Raw accelerometer and gyroscope measurements for a user carrying a handheld smartphone on 45 angle turn are shown in Figure 1. Step detection and step length estimation are greatly influenced by low-cost smartphone IMU sensors noise, unique gait pattern of users, variable walk speed, and phone carrying poses. The PDR is also susceptible to the accumulation of sensor drift error over long trajectories.

2.2. Experimental setup

The experiments were conducted on five participants with handheld android phone measuring accelerometer and gyroscope data. The setup consisted marked straight lines of pre-determined lengths and turns of 45°, 90°, 135°, 225°, 270°, 315° and 360 degrees. The participants were requested to

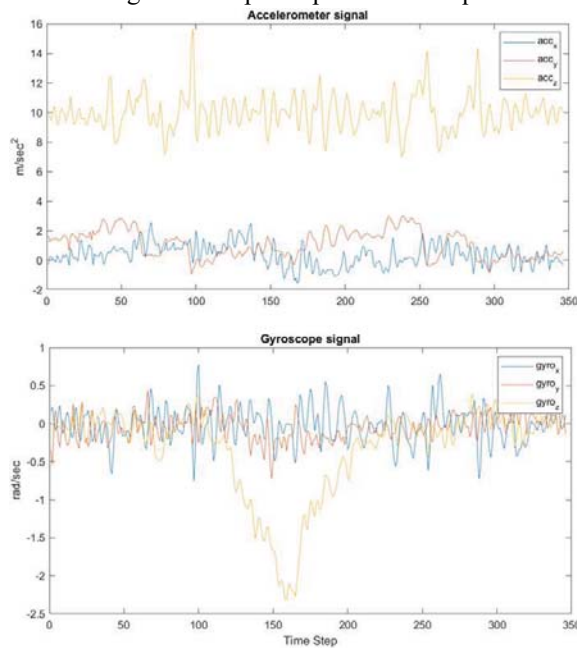


Figure 1: Raw smartphone IMU sensors signals for 45° turn

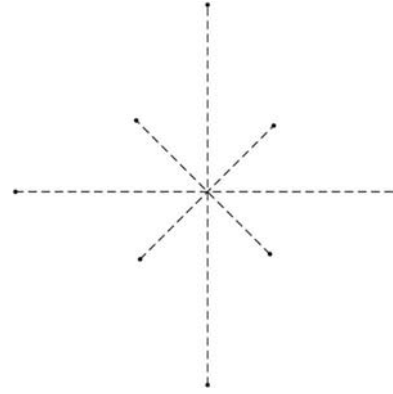


Figure 2: Marked experimental setup for model training

walk with handheld smartphone while; sensors measurements, starting position, ground truth trajectory and number of steps on each path were recorded for every trial. The participants were given a choice to opt their own trajectories on the marked map, as illustrated in Figure 2, for the model to learn parameters generalization over random class distribution. In addition, a dataset of 19 participants over two trajectories of 26 steps on average with one left and one right turn each as stated in [12] was included for model training and performance evaluation. Furthermore, sensor measurements for user handheld stationary position and miscellaneous activities such as texting, picking up and putting down the phone, and transitioning between sitting and standing to walk were also recorded. Ground truth labels for step length are calculated by dividing the length of the trajectory by the total number of steps taken to cover it, and turns were made between the end of previous steps and the start of the next step only. All sensor measurements were scaled to the range of -1 to +1, and segments of individual activities and steps were generated. These segments were randomly distributed by 75% and 25% to be used as training and test data preserving the overall proportion of activities into training and test sets.

2.3. System architecture

The proposed indoor positioning system consists of multi-stage activity classification, step length estimation, heading classifier and displacement accumulator blocks as shown in Figure 3. The system is designed to keep track of user's current position which is updated in displacement accumulator on every step and turning angle prediction by predecesing blocks. The activity classification block consists of single LSTM layer [13]. The step length estimation block comprises of sequence-to-sequence LSTM layer, another sequence-to-output LSTM layer followed by fully connected and regression layers. Heading classification block comprises of sequence-

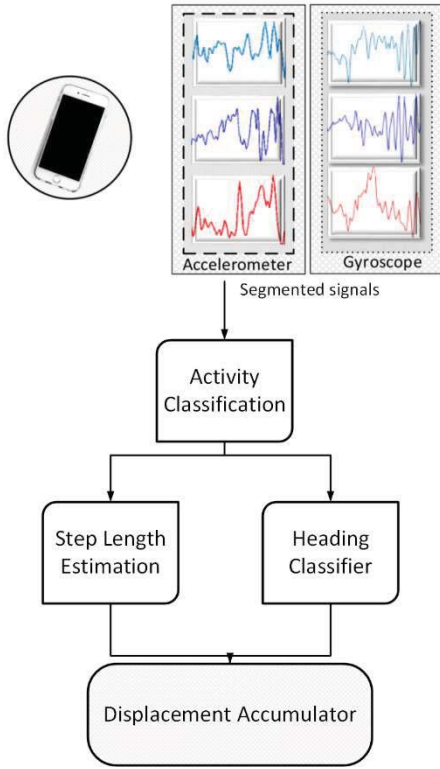


Figure 3: Architecture of deep learning based indoor localization system

to-sequence LSTM layer, another sequence-to-label LSTM layer followed by fully connected and classification layers. Activity classification, step length estimation and heading classifier are trained using activity, user step and user turn segments of smartphone accelerometer and gyroscope measurements provided at respective inputs. Ground truth activity labels, step length and turning or heading angle are also provided at output of respective blocks during training phase of deep network.

3. Performance evaluation

The performance of deep learning system was evaluated on test dataset, whereas, test data was not included during training of system. The activity classifier predicted 96.22% of activity segments correctly, breakdown of which is provided in Table 1. Relative error of N step segments was evaluated using;

$$error = \sum_{i=1}^N \frac{|y_i - \hat{y}_i|}{y_i} \times 100$$

Where, y_i is ground truth step length and $\hat{y}_i$ denotes predicted step length for i th step segment. Step length estimation block of proposed system resulted 4.77% of average error in estimated step length relative to ground truth step length. Heading classifier predicted with an accuracy of 71.42% on test set of heading direction change segments, it is the first deep learning-based heading classifier for indoor

localization to the best of our knowledge, Table 2 enlists the confusion matrix for each turning angle.

Table 1: Performance of activity classifier

Activity	Precision (%)	Recall (%)
Step	97.42	98.05
Turn	82.54	74.29
Stationary	96.69	99.15
Miscellaneous	89.47	85.00
Accuracy	96.22%	Error Rate 3.78%

Table 2: Confusion matrix of heading classifier performance

Output Class	45°	90°	135°	225°	270°	315°	360°
	4	1	0	1	0	0	0
	0	7	1	0	2	0	1
	1	2	3	0	0	0	0
	0	0	1	4	1	0	0
	0	2	0	0	8	0	0
	0	0	0	0	1	5	0
Turn	0	0	0	0	0	0	4
	45°	90°	135°	225°	270°	315°	360°
		Target Class					

4. Conclusion

This paper presented a deep learning method for the accurate estimation of a user in an indoor environment. It utilizes smartphone accelerometer and gyroscope sensors henceforth external hardware deployment is not required in contrast to other radio and non-radio based indoor localization techniques. The system employs automatic feature learning and is end-to-end trained; hence threshold or manual feature extraction and selection is not subjective rather data-driven. The proposed system is capable of estimating continuous step lengths by regression and is able to predict heading direction change between two consecutive steps for a finer resolution of turning angles instead of merely predicting right and left turns, which are improvements over existing deep learning system based on smartphone IPDR technique.

In future, we will investigate data-driven techniques for advancing heading and step length estimation in addition to the automatic segmentation of sensors signals.

Acknowledgments

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